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# INTEGRATED SIMULATION OF GRINDING AND FLOTATION APPLICATION TO A LEAD–SILVER ORE

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## ABSTRACT

*The paper outlines a procedure to simulate a grinding–flotation plant. Population balance models are used to describe the grinding, classification and flotation processes. In the grinding and classification models particles are characterized by their size, while in the flotation models particles are characterized by their size and mineral composition. The link between the grinding and flotation circuits is made by an empirical model that generates the mineral-size particle population from the size distribution of the grinding circuit product. The simulator is calibrated using plant data collected from the Peñoles-Fresnillo Pb–Ag concentrator in Mexico. © 1999 Elsevier Science Ltd. All rights reserved.*

**Keywords**
Grinding, Liberation, Classification, Flotation kinetics, Simulation.

## INTRODUCTION

The problem of defining the optimal fineness of a grinding circuit product to maximize the performance of the subsequent mineral separation steps is critical for mineral beneficiation plants. Laboratory tests provide part of the answer to this problem, but do not perfectly replicate the behaviour of continuous full scale operations. In practice, the optimal process tuning by planned experiments is a difficult task because of the inherent disturbances occurring in the process and the ore composition. Plant simulators are powerful tools for optimal tuning of grinding circuits in a disturbance-free environment. However, an operational link between flotation and grinding simulators is still missing, and the effects of changing the grinding circuit product size distribution on the separation stage performance are not completely taken into account by commercial simulators. Some studies were undertaken to incorporate particle composition and liberation phenomena in the grinding simulator, thus opening the door to the coupling of grinding and flotation simulators [e.g. 1,2,3,4,5,6,7]. However, the methods are still too complex to be easily used for optimization of integrated grinding/flotation circuits.

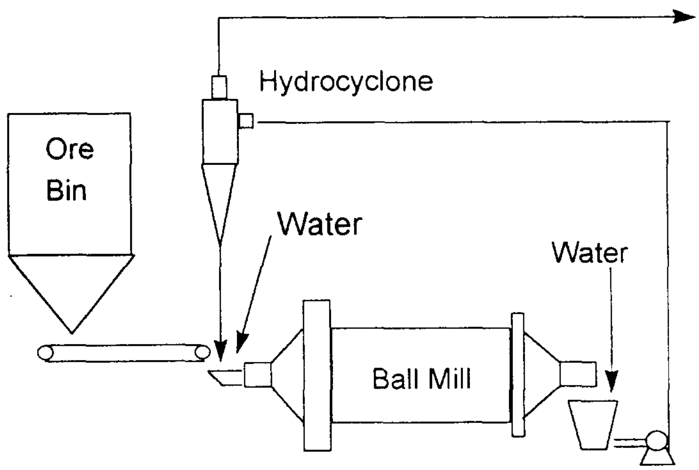
This paper proposes a simple approach to integrate grinding and flotation circuit simulation. It is based on an empirical model to predict mineral distribution within the size intervals of the flotation circuit feed. Some

researchers have used empirical relationships to estimate the chemical analysis as a function of particle size [7] and others have proposed to study the liberation of minerals by image analysis of polished sections [e.g. 1,3,4,5,6]. In this study an empirical model is used to correlate the cumulative mineral size distribution with the fine product particle size distribution [8].
The paper begins with a description of the Fresnillo lead–silver ore and concentrator. Then it outlines the procedure to build the grinding and flotation simulators, as well as the link between the two processes. Finally the results of the integrated simulation are compared to the plant data.

## PLANT AND ORE DESCRIPTION

The grinding–flotation simulator was calibrated using data from the Fresnillo plant, a property of the industrial group Peñoles, located in Northern Mexico. The concentrator processes 2200 tons/day of a Pb–Ag ore to produce 70 tons/day of a silver–lead concentrate.

The grinding circuit shown in Figure 1 consists of three parallel grinding and classification units. The fine products of the three grinding lines are mixed to feed the flotation process.




**Circuits no. 1 and no. 2.**

Ball mills: 10' 8" x 11'1220 HP.

Hydrocyclones 20''

**Circuit no. 3.**

Ball mill : 9.5' x 11' 1220 HP.

Hydrocyclone 20''

Fig.1 Grinding circuit.

Flotation of the Fresnillo ore is carried out in two steps, as shown in Figure 2a. The process is designed to selectively separate gold, silver and lead species from sphalerite, pyrite, arsenopyrite and non-sulphide gangue. The first circuit processes a silver high grade feed (900 g/t Ag). The tailings of the high grade circuit are classified and the fines are treated by a low grade circuit, while the coarse portion goes to regrinding and is returned to the flotation feed. In this work only the high grade circuit of Figure 2b with the regrinding stage is simulated.

The ore was characterized by X-ray and electron-scanning-microscope of the flotation feed samples. The main identified species are presented in Table 1 and the results of a liberation study are shown in Figure 3. The liberation was measured by a counting procedure. The total number of particles bearing the target mineral is firstly determined in an optical frame, then each particle is examined for further classification.

The selected discrimination classes were : liberated particles, binary particles (two minerals) and multiphase particles (more than two minerals). A mineral grain with less than two percent of any other mineral is defined as liberated. Sufficient optical frames were inspected to ensure that a minimum of one hundred particles of the target mineral were identified and classified.

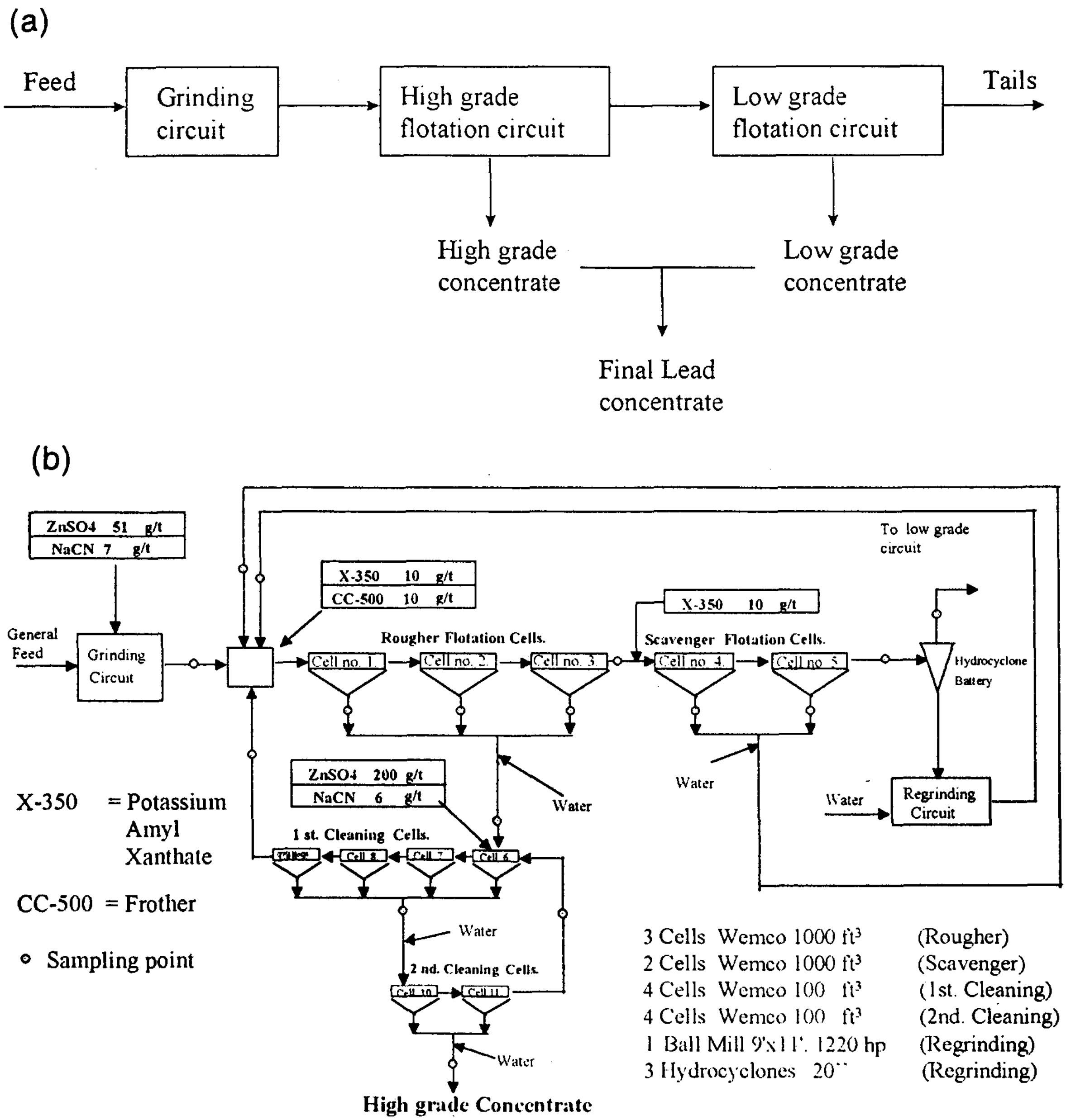



Fig.2 The overall grinding/flotation plant (a) and the high grade flotation circuit (b) including sampling points and reagents addition.

**TABLE 1 Mineral and metal composition of the ore**

| Mineral | Mineral assay | Metal | Metal Assay |
|---|---|---|---|
| Gold (Au) | 2.00 g/t | Au | 2.00 g/t |
| Pyrargyrite ($Ag_3SbS_3$) | 0.16% | Ag | 948 g/t |
| Galena (PbS) | 0.73% | Pb | 0.63% |
| Sphalerite (ZnS) | 2.01% | Zn | 1.28% |
| Pyrite ($FeS_2$) | 9.90% | Fe | 5.00% |
| Arsenopyrite (FeAsS) | 0.85% | As | 0.39% |
| Non-sulphide gangue | 84.35% | — | — |

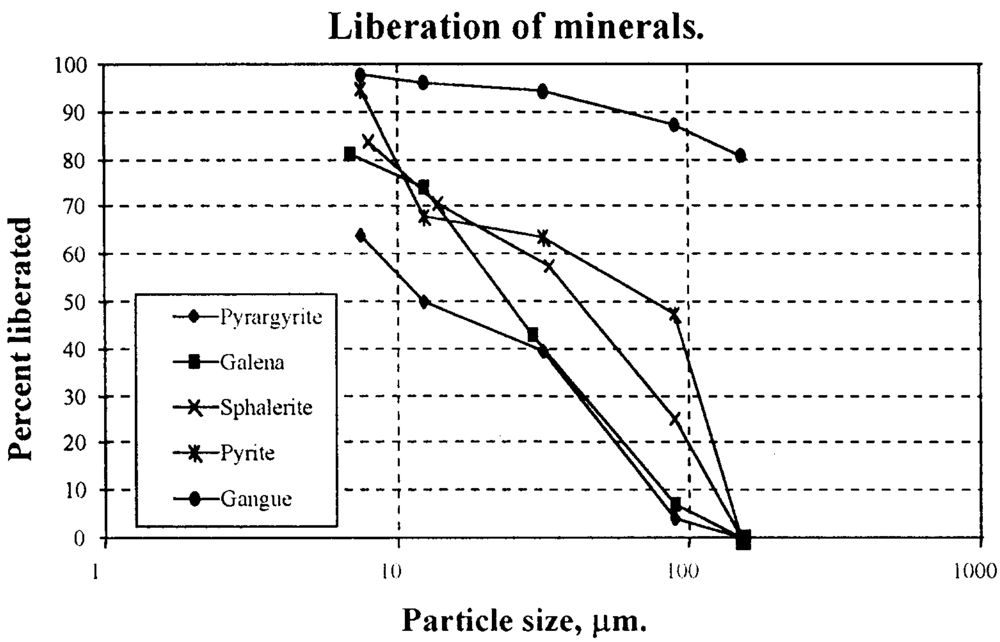



Fig.3 Mineral liberation as a function of particle size.

The results show that pyrargyrite exhibits a behaviour similar to galena in the coarse fractions, but, in the fines, galena tends to have a higher liberation. The non-valuable minerals such as pyrite and sphalerite have a higher level of liberation. Gold and arsenopyrite were not included in the liberation study because of their low content in the ore.

The following procedure is used to convert metal assays to mineral contents :

% $Ag_3SbS_3$ = $1.6734 \times 10^{-4}$ (g/t Ag)

% PbS = 1.1547 (% Pb)

% ZnS = 1.5688 (% Zn) (Sphalerite with 5 % iron)

% FeAsS = 2.1733 (% As)

% $FeS_2$ = 2.1481 [(% Fe) – 0.05(% ZnS) – 0.3430(% FeAsS)]

% Gangue = 100.00 – [% $Ag_3SbS_3$ + % PbS + % ZnS + % FeAsS + % $FeS_2$]

## MODELING OF THE GRINDING AND CLASSIFICATION PROCESSES.

### The ball mill

The phenomenological model used to represent the ball mill is derived from the population balance approach [9,10] based on the following first order batch grinding equation :

$$\frac{dm_i(t)}{dt} = -S_i m_i(t) + \sum_{\substack{j=1 \\ i>1}}^{i-1} b_{ij} S_j m_j(t), \qquad n \geq i \geq j \geq 1 \tag{1}$$

where $S_i$ characterizes the rate of breakage of size i particles (the selection function), $b_{ij}$ the fraction of fragments of size i produced by the breakage of size j particles (the breakage function) and $m_i(t)$ the ore mass fraction in the i-th interval.

Since the simultaneous estimation of the $b_{ij}$ and $S_i$ from industrial data leads to inaccurate parameters, the normalized breakage function was first estimated by batch laboratory tests and considered as scale independent to estimate the selection function from industrial data [11].

The breakage and selection functions were parametrized by the following equations :

$$\left\{\begin{array}{l} B_{i,j} = b_1\left(\dfrac{x_i}{x_j}\right)^{b_2} + (1-b_1)\left(\dfrac{x_i}{x_j}\right)^{b_3} \\ b_{i,j} = B_{i-1,j} - B_{i,j} \\ \ln(S_i) = \ln(s_1) + s_2 \ln(x_i) + s_3[\ln(x_i)]^2 + s_4[\ln(x_i)]^3 \end{array}\right\} \tag{2}$$

The coefficients $b_1$ to $b_3$ and $s_1$ to $s_4$ were determined using a least-squares estimation procedure applied to twenty one batch grinding tests of mono size feeds [11]. The estimated coefficients are : $b_1 = 0.4117$, $b_2 = 0.4818$, $b_3 = 4.5277$, $s_1 = 0.3104$, $s_2 = -0.4864$, $s_3 = -0.2837$, $s_4 = 0.0466$, which lead to the estimated breakage and selection functions shown in Figures 4 and 5.

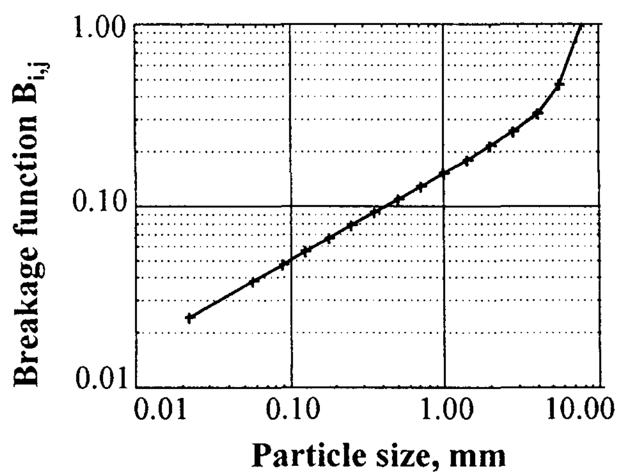



Fig.4 Normalized cumulative breakage function in the laboratory ball mill.

Figure 5 includes experimental values calculated directly from the curve $\ln[m_i(t)/m_i(0)]$ for the five different top size fractions used in the tests (7.77, 2.83, 1.00, 0.353 and 0.125 mm). For each top size fraction, values of the selection function are given at times 2, 4, 8 and 12 minutes, showing a variation of the top size kinetics as the grinding progresses. The continuous line represents the least-square estimates when the 21 test results are processed simultaneously for all the size fractions. For particles smaller than 1 mm, it shows lower rates of breakage when they are not the top size particles.

The ore Residence Time Distribution (RTD) required for the simulation is obtained by assuming that the ore behaves like water in the mill. Two tracer tests were performed in the plant using LiCl to evaluate the water RTD. The observed tracer response was modeled as two small perfectly mixed reactors of equal size in series with one large perfectly mixed reactor (mean residence times $\tau_1$ and $\tau_2$), as well as one plug flow reactor (mean residence time $\tau_p$) [12]. The results of Figure 6 show a good agreement between the experimental and simulated RTD for both tests.

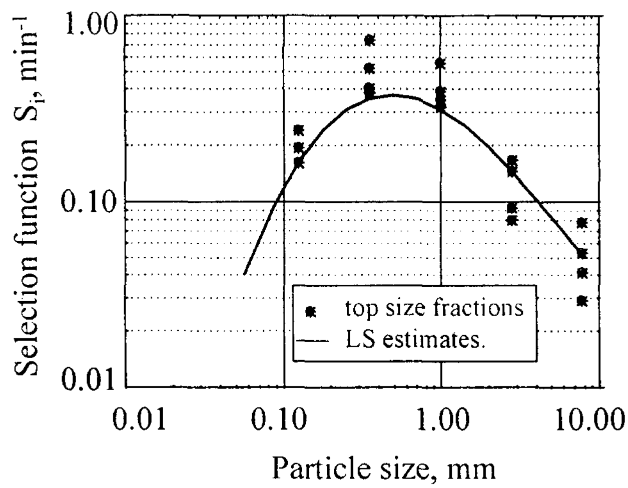



Fig.5 Selection function in the laboratory ball mill.

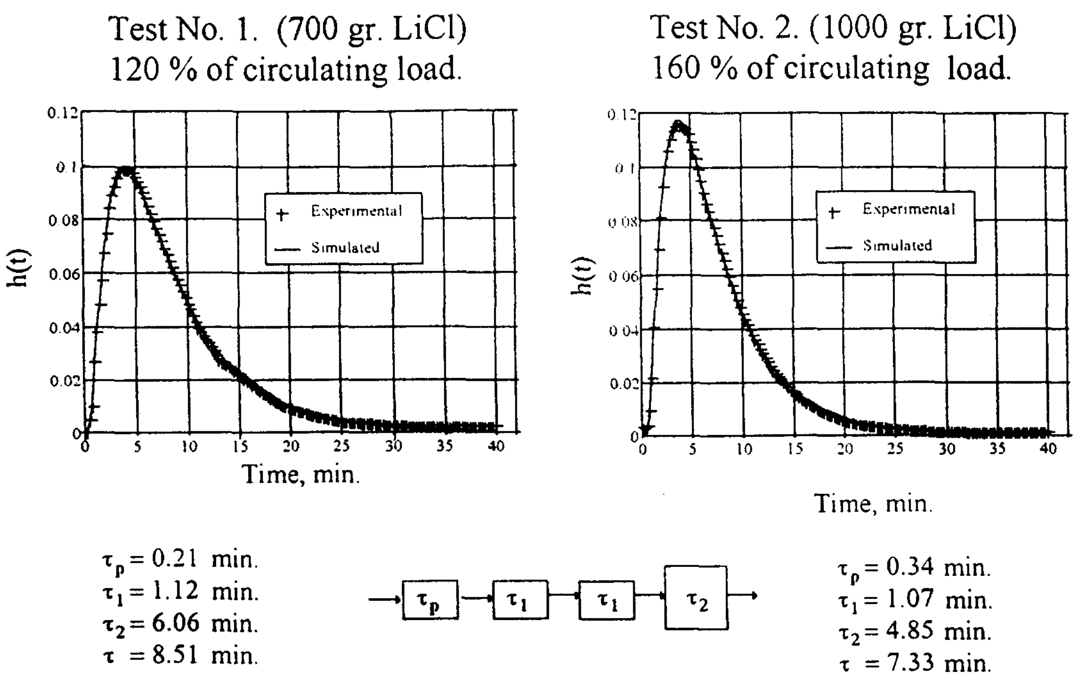



Fig.6 Comparison of experimental and modeled RTD of the industrial ball mill.

Once the breakage and RTD functions have been evaluated, the industrial scale selection function can be estimated. A sampling campaign was performed around the grinding circuit and the measured particle size distributions and slurry solids percent were reconciled using the Bilmat algorithm [13]. A least-squares technique was used to estimate the selection function from the data of ten sampling campaigns. Results lead to a second order functional form with the following coefficient values : $s_1 = 0.3864$, $s_2 = 0.8909$, $s_3 = -0.2299$ which produce the selection function of Figure 7a. In order to estimate the goodness of fit, the following relative error was calculated :

$$\mathrm{Er} = 100\sqrt{\frac{1}{(n-p)}\sum_{\text{tests}}\sum_{\text{fractions}}\left(\frac{m_i - \hat{m}_i}{\hat{m}_i}\right)^2} \tag{3}$$

where n is the total number of fractions in all the tests, p the number of parameters to estimate in the selection function, $\hat{m}_i$ and $m_i$ the estimated and experimental mass fractions in particle size classes at the ball mill discharge. A value of Er = 34 % was found, corresponding to the agreement between $\hat{m}$ and m shown in Figure 7b.

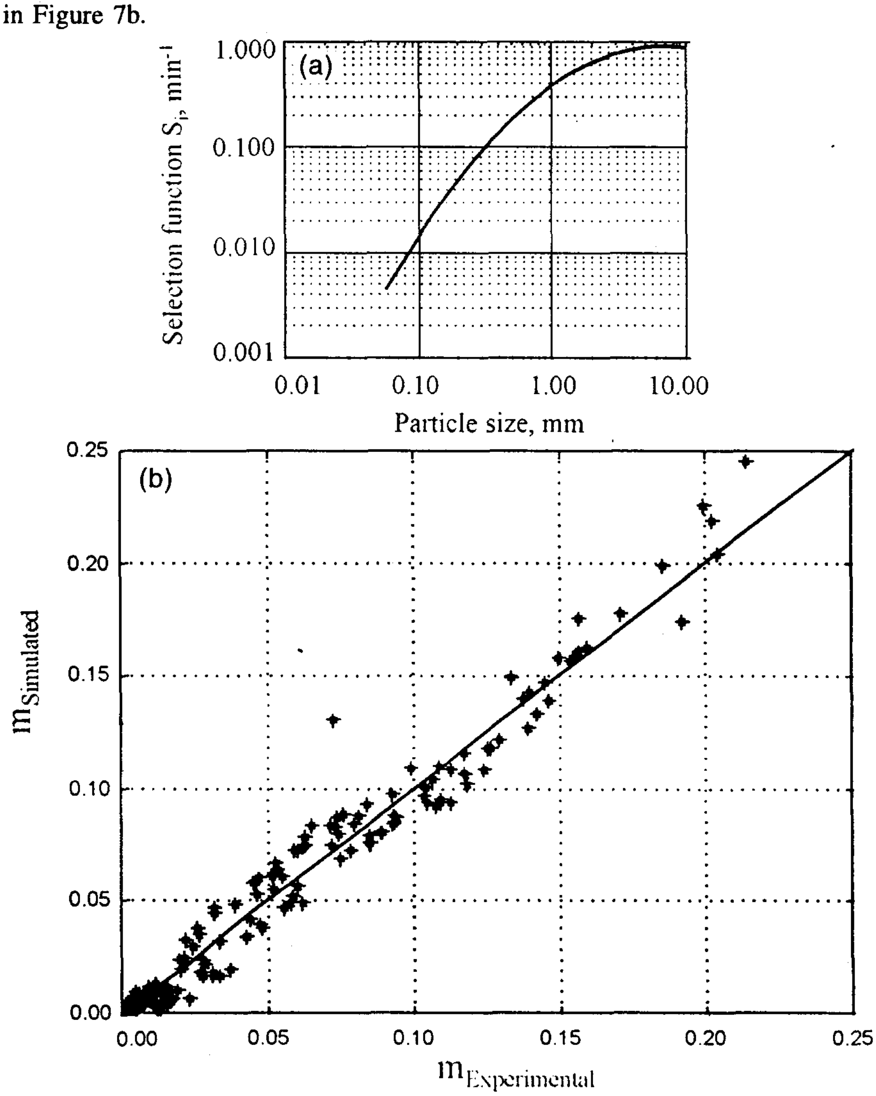



Fig.7 Selection function of the industrial ball mill (a) and comparison of experimental and simulated discharge size distributions (b).

**The hydrocyclone**

The Plitt's model with multiplicative correction factors for each equation is used to model the hydrocyclone [14]. The model was calibrated using the balanced data of ten sampling campaigns. Figure 8 compares the experimental and simulated corrected efficiencies for the second campaign. The correction factors applied to the Plitt's model are 0.81 for the corrected cut size equations, 0.59 for the sharpness factor equation, 1.05 for the pressure equation and 0.47 for the volume split equation.

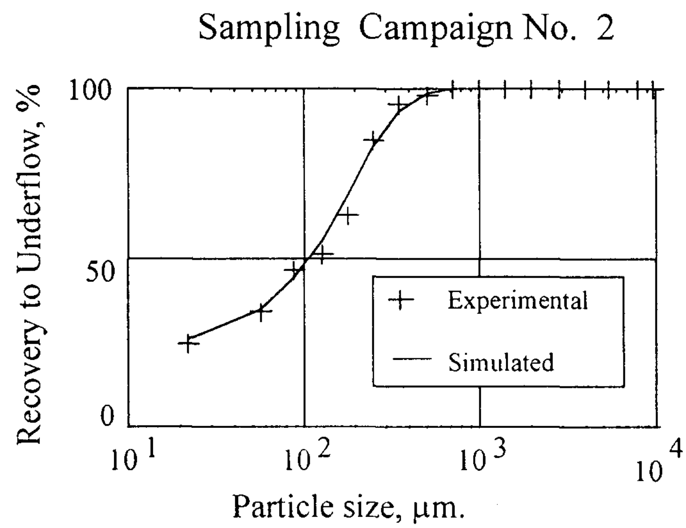



Fig.8 Hydrocyclone classification curve.

## MODELING OF THE MINERAL DISTRIBUTION IN THE FLOTATION FEED

Since flotation is a process sensitive to particle size and composition, its modeling requires a description of the minerals size distribution within the size intervals of the flotation feed. It would be ideal to completely characterize the particle population, specifying the mineral composition distribution and liberation within each particle size class. However, this would require a tremendous amount of experimental work, first to analyze the particles, then to predict the liberation phenomena at the grinding stage and finally to determine the flotation kinetics as a function of the particle size and composition. One way to avoid the modeling of complex particle flotation is to describe the flotation feed by the size distribution of the individual minerals, assuming that the ore behaves as a population of independent minerals. As shown in the ore description, minerals are not perfectly liberated ; however, flotation performances can be reasonably approximated by the behaviour of this equivalent population of particles.

To predict the flotation feed population of particles, the grinding model should be applied to particles of different sizes and compositions, thus requiring a definition of the classes and their grinding rate properties. Compared to the grinding of an homogeneous material, the number of parameters to describe the size and composition of the fragments increases drastically. This approach was proposed in the batch and continuous grinding modes for binary systems [1,2]. A simplification of this approach was also presented by Morrell and coworkers [6]. Constant breakage and selection functions and a simple classification model are assumed for a binary A–B system, defining three types of particles: liberated (A), unliberated (A–B) and bulk ore (B). As this approach still requires a lot of additional sample analysis when flotation and regrinding are included, and since it was anticipated that the estimation of individual S and B functions for the minerals would be extremely noise sensitive, an empirical approach was preferred.

It has been experimentally observed that the particle size distribution of the minerals is related to the overall size distribution [8]. For a complex sulphide ore, it has been shown that there exists a consistent relationship between the ore cumulative size distribution and the mineral cumulative size distribution. This is particularly obvious for minerals which systematically lead to finer size distribution than the ore. For a given grinding procedure, the relationship is fairly constant. However, it is not the same for open circuit grinding at the laboratory scale and for full scale closed grinding circuits. The different behaviour of the various minerals is attributed to the selectivity of grinding and classification processes. Hydrocyclones return denser material to the grinding circuit, thus explaining the finer size distribution of these denser minerals [14]. Also, grinding kinetics, as well as fragment distributions, may be different for the different minerals. In addition, there is preferential intergranular breakage which reveals the textural properties of the ore, for instance the finer grain size of the galena, in comparison to the gangue. The model proposed by Bazin [8] does not pretend to explain the observed behaviour, but is a potential tool for predicting mineral size distributions from ore size distribution and a simple method to link grinding and flotation processes without requiring a liberation process model. The Bazin's model uses the following third order polynomial to link ore and minerals size distribution :

$$MC_{ij} = a_{i,0} + a_{i,1}OC_j + a_{i,2}OC_j^2 + a_{i,3}OC_j^3 \tag{4}$$

where $MC_{i,j}$ is the mineral i cumulative percent passing size j and $OC_j$ is the ore cumulative percent passing size j.

The model was tested for the Fresnillo ore using the results of 10 different samples of the hydrocyclone overflow for different operating conditions of the mill. Table 2 and Figure 9 present the results of the model calibration by a least-squares method. It is clear from Figure 9 that the ore particle size distribution is controlled by the non-sulphide gangue size distribution, while all other minerals exhibit a finer size distribution, up to 35% passing. For gold, pyrargyrite, sphalerite and arsenopyrite, there are fewer fines in the last size intervals, down to 10 μm. Care should be taken if one is to extrapolate for particles finer than the experimental range. It is recommended, when possible, to use a subsieve method of size separation for defining the curve in the fine region.

**TABLE 2 Parameters of the mineral distribution model**

| Parameter | Au | $Ag_3SbS_3$ | PbS | ZnS | $FeS_2$ | FeAsS | Gangue |
|---|---|---|---|---|---|---|---|
| $a_0$ | −50.5171 | −57.2814 | −58.4442 | −39.0914 | 4.3405 | −47.5747 | 1.5292 |
| $a_1$ | 2.9851 | 3.8895 | 4.8778 | 2.6902 | 1.0863 | 3.4168 | 0.9019 |
| $a_2$ x$10^3$ | −19.8171 | −34.3067 | −53.0826 | −12.4260 | 2.3950 | −29.4616 | 0.8622 |
| $a_3$x$10^5$ | 5.0174 | 11.1402 | 20.1491 | −0.5667 | −3.6919 | 10.0506 | −0.0338 |

## MODEL OF THE FLOTATION PROCESS.

The model used for a flotation cell is based on the following assumptions :

1. Flotation and entrainment are responsible for the transport of particles from the pulp to the concentrate.
2. Perfect mixing of water and particles occurs within the cell.
3. The entrainment of pulp into the concentrate is related to the water in the concentrate, through an entrainment factor depending upon the particle mass.
4. The particle flotation rate is assumed to be a first order process which depends upon mineral species and particle size.

5. The transfer of particles at the pulp–froth interface is lumped into the flotation and entrainment

5. The transfer of particles at the pulp–froth interface is lumped into the flotation and entrainment phenomena.

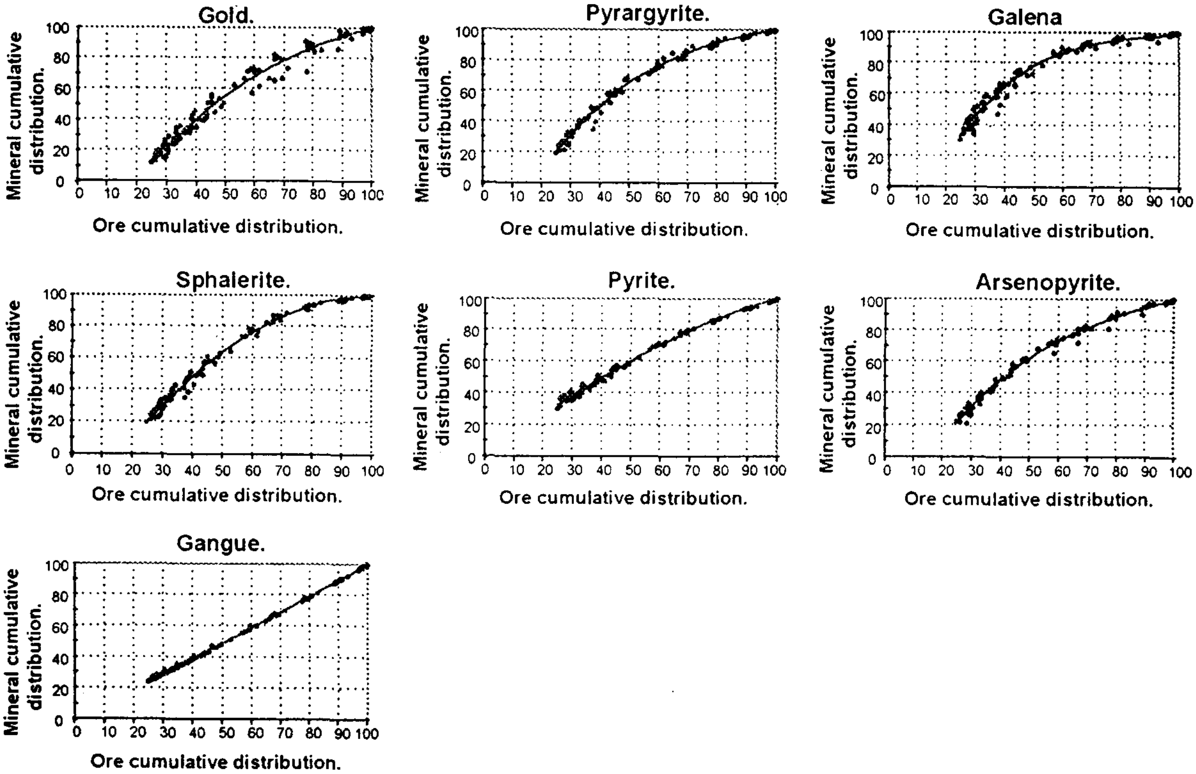



Fig.9 Comparison of experimental and modeled cumulative mineral size function of the ore cumulative particle size distribution.

Particle recovery by flotation is induced by particle hydrophobicity. The reaction between the collector and the free sulphide surfaces produces hydrophobic particles which, in contact with the air bubbles in the pulp, form stable air–mineral aggregates. Eventually, the agitation in the cell can detach the minerals from the bubbles and only the most stable aggregates survive the cell turbulence and pass to the froth zone. Some loaded bubbles collapse in the froth, part of the minerals returns to the pulp and the rest goes to the concentrate. The combination of these three phenomena leads to the concept of net flotation of particles to the concentrate which is used in the model. A flotation kinetic constant and a mean residence time are then used to simulate particles flotation [e.g. 15,16].

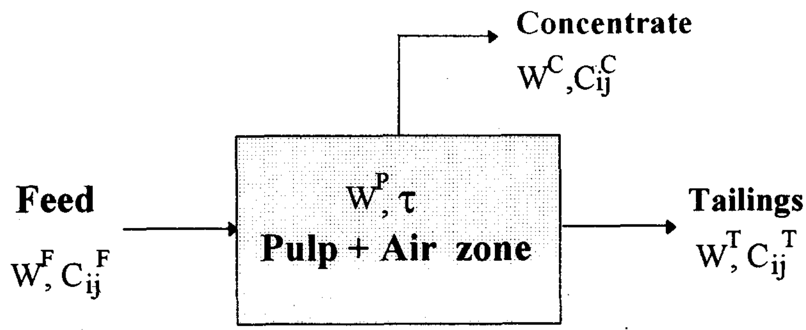
The loaded bubbles rising from the pulp to the froth zone are covered with a film of pulp, probably thicker at the rear part of the bubbles. This phenomenon is responsible for the pulp entrapment in the froth by the collective ascension of the bubbles. Because of bubble coalescence and hydrophilic particle detachment in the froth, some slurry is returned back to the pulp zone. The balance of these two phenomena (drainage and entrainment) is the net entrainment of particles into the concentrate which is used in the model.

The entrainment is a process which involves not only the non-sulphide gangue, but all the minerals in the pulp. The evaluation of entrainment and kinetics as contributions to the mineral recovery in the concentrate has been studied by different authors using laboratory batch tests [e.g. 17,18]. The usually accepted model relates the recovery of any particle to its mass and to water recovery [19].

Under these assumptions, the mass balance equations for the particles of mineral i into the size fraction j are (see notations in Figure 10) :

$$W^F C_{ij}^F - W^T C_{ij}^T - W^C C_{ij}^C = 0 \tag{5}$$

where the species concentrations are expressed as the mass ratio of particles ij to water. Using water as a reference phase is rather unusual, but is more convenient since it simplifies the mass balance equations.




$W^F, W^T, W^C$ = water flowrates in feed, tailings and concentrate.
$W^P$ = water hold-up.
$C_{ij}^F, C_{ij}^T, C_{ij}^C$ = mass of particles « ij » per unit mass of water.
$\tau$ = tailings mean residence time.

Fig.10 Distribution of water and solids in a flotation cell.

The water mass balance equation is :

$$W^F - W^T - W^C = 0 \tag{6}$$

The particles flowrate into the concentrate is the sum of the entrainment and flotation phenomena :

$$W^C C_{ij}^C = K_{ij} W^P C_{ij}^T + e_{ij} W^C C_{ij}^T \tag{7}$$

where $K_{ij}$ and $e_{ij}$ are respectively the flotation rate constant and the entrainment factor of the mineral i in size interval j. The entrainment is modeled using the following empirical representation of an s-shape curve (look ahead to Figure 12a) :

$$e_{ij} = \exp\left[-c_1\left(\frac{M_{ij}}{c_2}\right)^{c_3}\right] \tag{8}$$

where $c_1$, $c_2$ and $c_3$ are the coefficients to be estimated and $M_{ij}$ is the mass of the entrained particles ij.

The water holdup and the water behaviour in the cell are not modeled. Instead it is assumed that the mean residence time $\tau$ and the water split factor $\alpha$ are known from the experimental data :

$$\tau = \frac{W^P}{W^T} \tag{9}$$

$$\alpha = \frac{W^C}{W^T} \tag{10}$$

The mean residence time $\tau$ was measured by two tracer tests in the rougher and scavenger cells. Results show that the effective pulp volume in the cell is 78 % of the total volume (Figure 11).

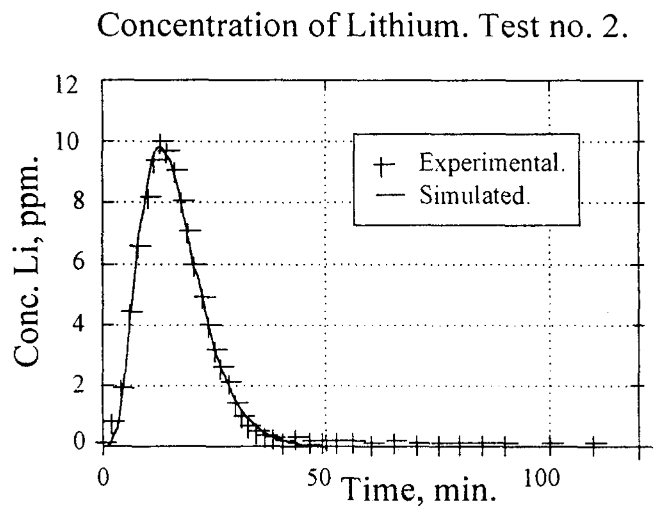



Fig.11 Simulated and experimental concentrations of tracer in the cell 5 tailings water (Test no. 2).

From the above equations one can express the fraction $\varphi_{ij}$ of particles ij remaining in the tails as :

$$\varphi_{ij} = \frac{W^T C^T_{ij}}{W^F_{ij} C^C_{ij}} = \frac{1}{1 + \alpha e_{ij} + \tau K_{ij}} \tag{11}$$

which for each cell k in the rougher–scavenger bank is :

$$\varphi^k_{ij} = \frac{1}{1 + \alpha_k e_{ij} + \tau_k K_{ij}} \tag{12}$$

assuming that the entrainment factor and flotation rate coefficient are constants along the flotation bank.

In order to calibrate the model, a sampling campaign was performed around the high grade circuit. The sampling points are indicated in Figure 2b. All the collected samples were separated into fractions by sieve analysis and the −325 mesh fraction was processed in a cyclosizer. All the size fractions were assayed for gold, silver, lead, zinc, iron and arsenic. The raw data were reconciled using the Bilmat algorithm [20,13] and metal assays converted into minerals using the previously described procedure.

The results of mineral recovery size by size as a function of mean residence time and water recovery in the rougher and scavenger banks confirm that flotation and entrainment contribute to particle recovery and should be considered in the model.

The estimation of the model parameters $c_1$, $c_2$, $c_3$ and $K_{ij}$ from reconciled data is based on a two-step least-squares procedure. First, the coefficients $c_1$, $c_2$ and $c_3$ are estimated from the gangue behaviour. Then, the $K_{ij}$ are estimated from the behaviour of the various minerals, and the least-squares criteria are calculated using :

$$J_i = \sum_{k=1}^{5}\sum_{j=1}^{n}\left(R_{ij}(k) - \hat{R}_{ij}(k)\right)^2 \tag{13}$$

where $R_{ij}(k)$ and $\hat{R}_{ij}(k)$ are respectively the experimental and modeled cumulative recoveries of particles ij at cell k. $\hat{R}_{ij}(k)$ is given by :

$$\hat{R}_{ij}(k) = 1 - \varphi_{ij}^{(1)}\varphi_{ij}^{(2)}\cdots\varphi_{ij}^{(k)} = 1 - \prod_{\ell=1}^{k}\varphi_{ij}^{(\ell)} \tag{14}$$

where the $\varphi_{ij}$'s are evaluated by the current values of $e_{ij}$ and $K_{ij}$ while $\alpha_k$ and $\tau_k$ are obtained experimentally.

In the first estimation step, Criterion 13 is minimized with respect to $c_1$, $c_2$ and $c_3$, i being the gangue and assuming that it does not float at all ($K_{gangue,j} = 0$). Then, in a second step, $J_i$ is sequentially minimized for each mineral with respect to $K_{ij}$, $e_{ij}$ being calculated from the first step values of $c_i$, $c_2$, $c_3$ and Equation (8), where $M_{ij}$ are calculated assuming spherical particles.

The results for entrainment are presented in Figures 12a and 12b ($c_1$ = 4.8; $c_2$ = 0.09; $c_3$ = 0.12). The flotation rate constants are presented in Figure 13. Gold, galena and pyrargyrite exhibit, as expected, higher rates of flotation compared to sphalerite, pyrite and arsenopyrite. For all the minerals the flotation rate increases as particle size decreases. The maximum rate of flotation is around 10 μm, a surprisingly low value. Particles finer than 10 μm exhibit a flotation rate comparable to that of the 50 μm particles. The rate of flotation of the 100 μm particles is less than a fifth of the maximum flotation rate, while the flotation rate of the 200 μm particles is almost zero.

## VALIDATION OF THE GRINDING–FLOTATION SIMULATOR

The grinding circuit of Figure 1, followed by the flotation circuit of Figure 2b, is simulated in an integrated way, by linking the grinding circuit product to the flotation feed with the empirical model presented above. The grinding circuit is simulated by the iterative modular–sequential approach using the recycle stream as a torn stream, initialized at a zero flowrate. The simulated ore size distribution is used to predict the cumulative mineral size distribution (Equation 4) and to describe the feed to the flotation circuit. Again, the modular sequential approach is used and all the recycled streams are torn and initialized at zero flowrate values.

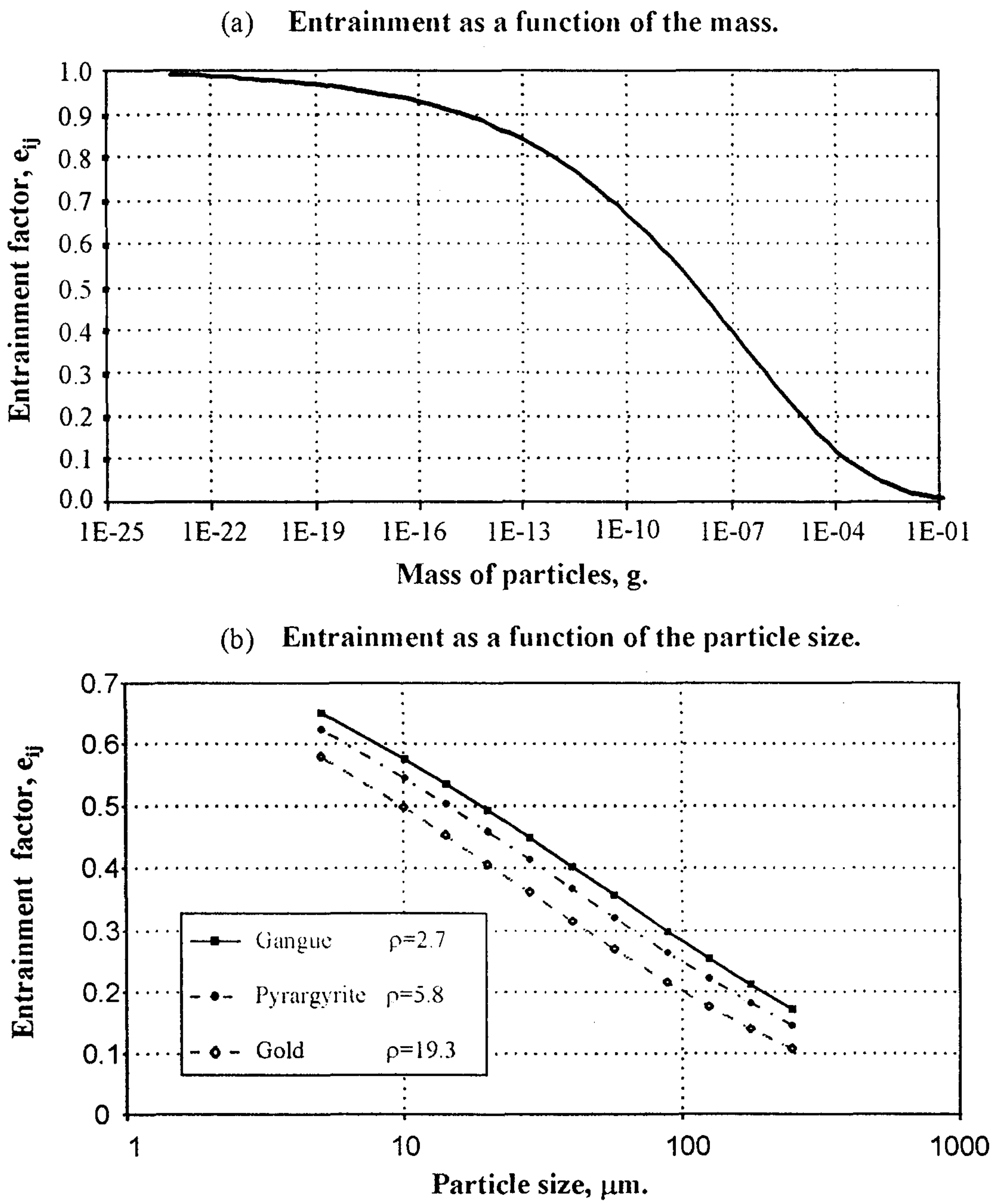



Fig.12 Entrainment factor as a function of particle mass (a) and particle size (b).

Each cell of the rougher–scavenger bank is simulated using the model parameters described in the previous sections. The concentrate percent solids of each cell is set to the target values and the cell is iteratively simulated starting with an initial value of the mean residence time $\tau_k$ , updated at each iteration up to convergence. For this purpose the pulp volumetric holdup is assumed to be constant. When a cell is converged, the program starts the simulation of the next one.

The data collected around the cleaning stages are insufficient to allow the estimation of the flotation rate constants and entrainment factors for these units. The simulation of the cleaning stages is performed assuming constant separation efficiencies for each class ij of particles for each cleaning stage k (k = 1, 2) :

$$s_{ij}^{(k)} = \frac{W^C C_{ij}^C}{W^F C_{ij}^F} \tag{15}$$

where W and C are the water flowrates and particle ij mass fractions of the cleaning stage k.

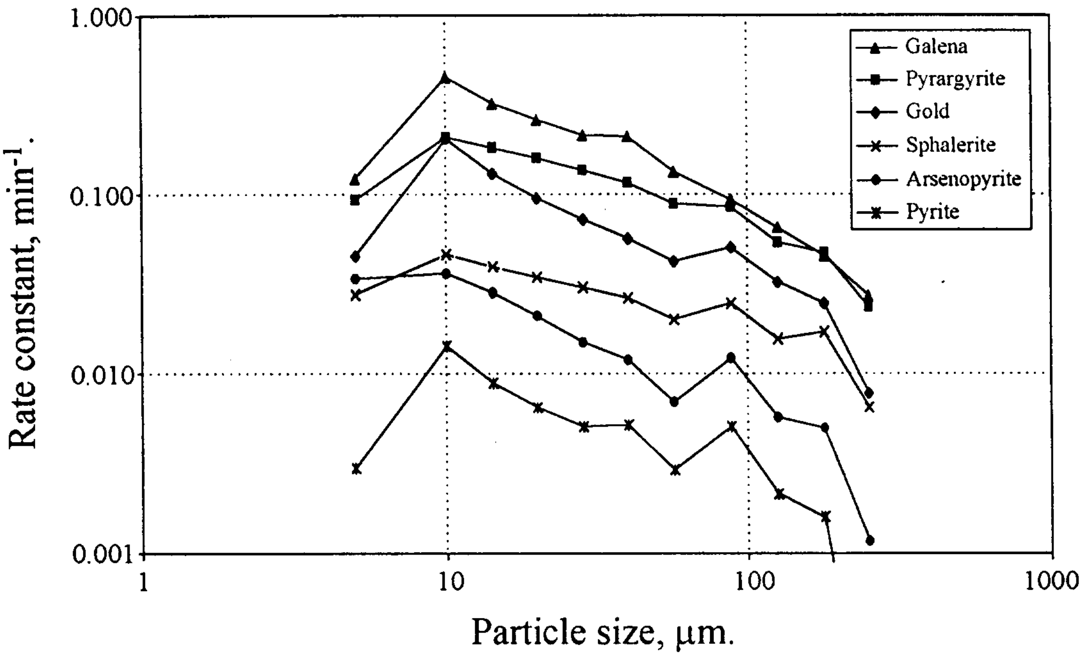



Fig.13 Flotation rate constant ($min^{-1}$) as a function of particle size.

For the regrinding circuit the simulation was performed assuming that the minerals in the first hydrocyclone have given classification curves. The breakage function used is the same as the one estimated for the simulation of primary grinding mill, and the selection function of the ball mill was scaled using a relationship involving the mill diameter [10]. The size-by-size mineral distribution in the regrinding product was also estimated using the model proposed by Bazin [8] and calibrated for the Fresnillo ore.

The simulator generates a large amount of results for the 23 simulated streams. To simplify the presentation of the results, the comparison of the experimental and simulated values will be performed only for the following variables :

a) Particle size distribution of the grinding circuit fine product ;
b) Size-by-size chemical assays of the grinding circuit fine product ;
c) Final concentrate grade and recovery by species.

These results are presented in Figures 14 to 16.

Figure 14 shows the results for the grinding circuit products corresponding to two significantly different operating conditions (221 and 439% for the circulating load ratio). The fines in the hydrocyclone overflow exhibit a bias probably due to the calibration range, since the breakage and selection functions were estimated with grinding tests from −9.51 + 6.35 mm to −149 + 105 microns and the classification curve for the hydrocyclone was calibrated in the range of −9.51 + 6.35 mm to −44 microns.

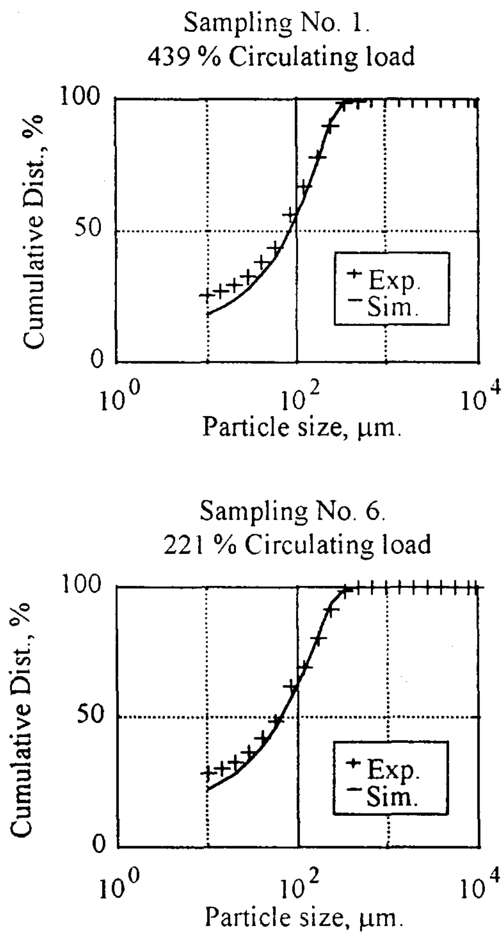



Fig.14 Particle size distributions in the grinding circuit product.

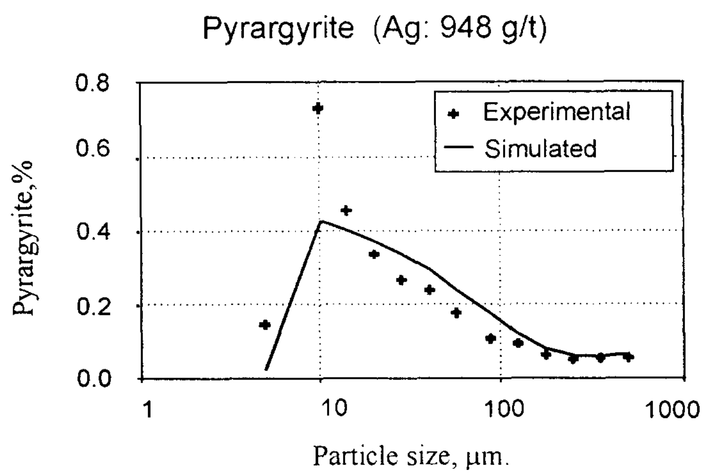
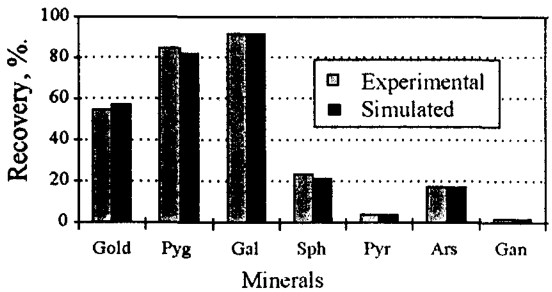



Fig.15 Comparison of experimental and simulated pyrargyrite grades in the hydrocyclone overflow.

Figure 15 compares the chemical assays as a function of the particle size in the fine product (948 g/t of silver in the feed). A significant difference is observed for the 10 μm particles in the chemical assays prediction after grinding. This is believed to be caused by the difficult problem of linking sieve and cyclosizer analysis in the data filtering procedure.



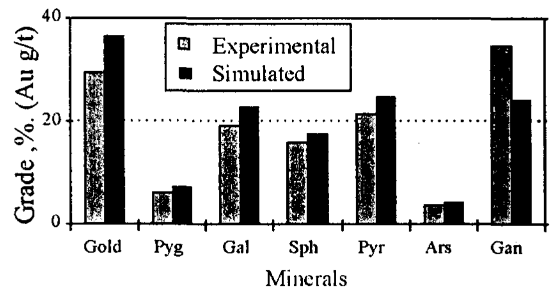




Fig.16 Comparison of experimental and simulated grades in the high grade concentrate.

The prediction of metal recoveries in the final concentrate (Figure 16) is in good agreement with the experimental results. The simulator correctly predicts the grade tendency with more deviations for the gold and gangue assays, which are inherently less accurate. Another source for the differences between simulation and experimental data is the assumption of perfect liberation in the flotation model.

Overall, the simulation results are quite satisfactory considering the large number of steps in the process of industrial data acquisition and experimental data processing, and the number of assumptions made for preparing the model. The most critical points might be :

- the fine particle analyses and the problem related to linking sieve and cyclosizer analysis for multimineral materials with large differences in specific gravity.
- the empirical approach used to correlate mineral size distribution and ore size distribution, without referring to the liberation degree which, at least, is assumed to be constant.
- the assumption that the minerals in the flotation circuit behave as independent particles, thus again assuming perfect liberation or ,at least, constant liberation degree.
- the assumption that cleaning stages produce constant separation efficiencies.
- the absence of a model of particle hydrophobisation to take into account the role of the reagents in the flotation circuit.

## CONCLUSION

This study offers a novel approach to link grinding and flotation simulators. The proposed relationship between the ore and minerals size distributions is empirical, but consistently observed for different ores. The approach has been validated for a silver–lead ore processing circuit. The results are considered quite encouraging, taking into account the complexity of the simulated plant which includes many recycling streams and a regrinding circuit, and the complexity of the ore for which the significant flotation behaviour is mainly in the fine size range. The main difficulties were encountered in the raw data processing, which required various steps including procedures to link different particle size ranges and to reconcile data with mass balance constraints.

The integrated grinding/flotation simulator has great potential for studying the optimal arrangement and tuning of grinding circuit in order to maximize the economic performance of the flotation process.

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